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# Electrodeposition of magnesium from halide melts—charge transfer and diffusion kinetics

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**Abstract**—Electrodeposition of magnesium from molten $MgCl_2$ and $MgCl_2$–$MgF_2$ (78–22 mol%) mixtures was studied by electrochemical techniques. The process was found to be quasi-reversible with a cathodic electrochemical rate constant for the total charge transfer reaction of about $10^{-3}$ cm/s at 780°C. Underpotential deposition of an adsorbed layer of reduced Mg(II)-species was observed during cyclic voltammetry and potential step measurements. The electrode capacitance was determined from galvanostatic pulses and found to depend on the applied current density. This has been explained by a pseudo capacitance due to the formation of an adsorbed layer. At potentials positive to nucleation a diffusion controlled process was identified. The determined concentration of the diffusing species, as well as the fact that Mg(II) species are the only cations in the system, indicate that the process is diffusion of dissolved metal (the product) from the electrode interface to the electrolyte. © 1997 Elsevier Science Ltd. All rights reserved.

*Key words:* Halide melts, electrodeposition, magnesium, kinetics, diffusion.

## INTRODUCTION

Most of the magnesium produced today is extracted by molten salt electrolysis. The high specific energy consumption (12–16 kWh kg) has been a motivation to improve the process with respect to energy and current efficiency. In order to develop the existing and future processes a detailed understanding of the different electrode reactions is necessary.

Cathodic deposition of magnesium from various chloride melts on different substrates has been studied by several authors [1–5]. Most of the electrochemical measurements in $MgCl_2$ containing melts reported in the literature have been performed in dilute solutions of Mg(II)-species, typically 0.1–10 wt% $MgCl_2$.

The cathode process in dilute solutions has been found to be controlled by diffusion of the reactant. Alloy formation has been observed when the electrode material was platinum, as reported by Tunold [1] and Duan *et al.* [3]. The rate constant of the charge transfer process on an Mg–Ni electrode in molten NaCl–$CaCl_2$–$MgCl_2$ has been reported by Tunold to have a value of about 0.01 cm/s. The author also reported the formation of a monolayer on iron electrodes, occurring at potentials approximately 100 mV positive to Mg deposition.

The advantage of using a melt with Mg as the only cation is that codeposition of other metals, such as Na and K, is avoided. Another simplification with the pure system is that the reactant is not transported by a diffusion process, obscuring other electrode processes. Nevertheless, only a few investigations in pure $MgCl_2$ have been reported to the knowledge of the authors.

The kinetics of a liquid magnesium electrode in molten $MgCl_2$ has been determined by combining *ac*-impedance spectra and galvanostatic relaxation transients by Kisza *et al.* [6]. These authors concluded that the charge transfer is a two-step process and that the rate determining step is the reduction of an Mg(II) species to an Mg(I) species whereas the second charge transfer is very fast. The charge transfer processes were supposed to be preceded by a slow chemical dissociation of polynuclear magnesium complexes.

Some interesting features observed during cathodic deposition of magnesium on glassy carbon electrodes in molten $MgCl_2$ have previously been reported by the present authors [7]. The deposition process was found to be unstable under certain circumstances.

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The instability was observed as an oscillating current during potentiostatic deposition of magnesium on glassy carbon.

Raman spectroscopic studies of molten magnesium chloride and alkali chloride–magnesium chloride mixtures have been reported by Balasubrahmanyam [8], Capwell [9] and Huang and Brooker [10]. There is a general agreement that the molten magnesium chloride retains the polynuclear structure in the solid state to a certain extent also in the liquid state. Huang and Brooker [10] observed peaks which were assigned to tetrahedral $MgCl_4^{2-}$ species also in pure $MgCl_2$.

The reported measurements are part of a more extensive study of the cathode process in pure molten $MgCl_2$, $MgCl_2$–$MgF_2$ (78–22 mol%) mixtures and alkali chloride melts with $MgCl_2$.

## EXPERIMENTAL

Anhydrous $MgCl_2$ was prepared by drying magnesium hexahydrate (Merck, p.a.) in HCl(g) during slow heating to 450°C. The dried $MgCl_2$ was subsequently distilled under vacuum ($P < 0.1$ Pa). $MgF_2$ (Merck, Optipur) was dried under vacuum with the temperature slowly increasing up to 600°C. The working electrodes were unsheathed rods (∅: 3 mm, h: 2–5 mm) of glassy carbon (Tokai Carbon, GC30S), iron (Johnson Matthey, Specpure) or molybdenum (99.97%) connected to graphite rods. The area of the working electrodes was determined separately in each experiment by changing the depth and area of immersion and determining the corresponding current change. The area determined by this method was in one experiment compared with the one found by visual inspection of the electrode surface (after use). In addition the current density response of this electrode was directly compared with the equivalent response of an electrode sheathed in boron nitride. The agreement was within 10%. In some measurements, an Mg working electrode, formed *in situ* by depositing magnesium on an Mo wire placed inside an oxide tube, was applied. The reference electrode was magnesium deposited on an Mo wire placed inside an MgO tube, while the counter electrode was a rod of graphite (Johnson Matthey, Spectrotech) or glassy carbon (Tokai Carbon, GC30S/GC20S). The temperature was measured by a Pt–PtRh (10%) thermocouple. All handling of the salts was carried out in a glove box with dry argon atmosphere. During measurements the atmosphere in the experimental cell was also argon which was dried by magnesium perchlorate and deoxygenated by titanium sponge before entering the cell.

Measurements were performed at temperatures ranging from 725°C to 780°C in the pure $MgCl_2$ system, while the temperature in the $MgCl_2$–$MgF_2$ mixture varied between 647°C and 720°C.

The electrochemical measurements were carried out using a PAR Model 173 Potentiostat-Galvanostat connected with a PAR Model 175 Universal programmer. The transients and voltammograms were recorded using an XY-recorder (Yokogawa 3023) or an oscilloscope (Nicolet 410).

## RESULTS AND DISCUSSION

The results presented in this paper are grouped in two parts, charge transfer kinetics and diffusion. The kinetic data were mainly obtained from current transients acquired with a time resolution of 1 μs. The mass transfer controlled process was investigated by potential step and voltammetric measurements, and the data were obtained at times ranging from 50 ms to several seconds.

### Charge transfer kinetics

The electrode kinetics was determined on electrodes of iron, molybdenum and glassy carbon in molten $MgCl_2$ and on glassy carbon and magnesium in molten $MgCl_2$–$MgF_2$. The obtained results are discussed in terms of the electrode materials.

*Electrodes of iron, molybdenum and glassy carbon.* A typical voltammogram obtained on glassy carbon (GC) electrodes in pure molten $MgCl_2$ is given in Fig. 1. Similar voltammograms were obtained in molten $MgCl_2$–$MgF_2$. The magnesium deposition is observed as wave (A). By reversing the potential scan, peak (B), which is typical for stripping of a deposit, is recorded. The chlorine evolution is observed as wave (C). At potentials slightly less positive than the chlorine evolution one notices a shoulder, which is probably due to the anodic formation of $CO/CO_2$ from an oxide species dissolved in the electrolyte [11]. The peaks D and D′ are most likely due to formation and dissolution of an adsorbed layer of reduced magnesium species equivalent to an underpotential deposition (UPD). The charge of the cathodic prepeak (D) corresponded to 1–3 monolayers while the charge corresponding to the anodic peak (D′) in general was somewhat higher and typically amounted to 3–8 monolayers. This means that the charge involved in the UPD increased during normal

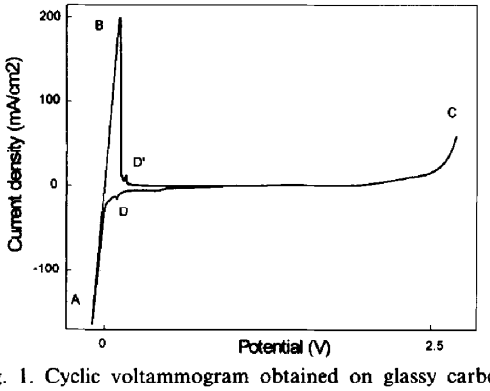



Fig. 1. Cyclic voltammogram obtained on glassy carbon in molten $MgCl_2$ at 725°C, sweep rate 100 mV/s, $E_{init} = 900$ mV.

deposition of Mg. The nucleation–growth–collision model for two dimensional monolayer formation predicts a $v^{1/2}$ dependency of the peak cd at low sweep rates whereas this changes to a linear dependency of $v$ at higher sweep rates [12]. Our results seem to exhibit the same type of behaviour. The difference between the peak potential of the underpotential deposition on glassy carbon and the potential of the magnesium reference electrode was found to be 50 to 100 mV, which corresponds to an energy difference of 10 to 20 kJ mol. The peak potential was found to have a negative shift with increasing scan rate. Such underpotential deposition has also been observed on metallic electrodes (Fe, Mo), but the observed peaks (due to the UPD process) were less reproducible on these electrodes than on glassy carbon electrodes. This could probably be due to problems related to removal of all magnesium deposited in the UPD process between each potential scan. There also always exists the possibility of a very small alloy formation with the metallic substrates, even if molybdenum is not known to alloy with magnesium, whereas iron alloys to an insignificant extent. This implicates that the monolayer formation depends on the electrode material. As has been reported, the cathodic charge associated with the voltammetric peaks for the UPD indicates that several layers are formed. A similar feature has been observed in studies of the formation of silver on platinum electrodes in aqueous solution [13].

The rate of the charge transfer on the different electrode materials was evaluated from potentiostatic current transients. By plotting the current transients vs $t^{1/2}$ and extrapolating the initial linear region to zero time, one may calculate the potential dependent cathodic rate constant using [14]:

$$i_{t=0} = nFk_c c_b, \tag{1}$$

where $i_{t=0}$ is the current density (cd) obtained by extrapolation of the current transient to zero time and $k_c$ and $c_b$ are the potential dependent cathodic rate constant and the bulk concentration of reacting species, respectively. In equation (1) one assumes that the rate of the back reaction is negligible during the very initial part of the potential step and hence that the value of $i_{t=0}$ obtained by extrapolation to the reversible $Mg^{2+}/Mg$ potential should be equal to the exchange current density ($i_0$).

Typical current transients obtained on GC in molten $MgCl_2$ and $MgCl_2$–$MgF_2$ are given in Fig. 2 (A) and (B) respectively. The cathodic current transients plotted as a function of $t^{1/2}$ are also given in each of the two figures. From Fig. 2 (B) ($MgCl_2$–$MgF_2$ mixture) one easily observes phase formation in the initial part of the transients. This phase formation obviously affects the determination of the exchange cd and hence the calculation of the rate constant since the electrode will be partly, or fully, covered with a layer of reduced Mg species. The difference in behaviour for this process in the two melts is probably due to differences in interfacial energies, the mixed system possessing a higher surface energy than pure $MgCl_2$ [15]. The values of the cathodic rate constant, $k_c$, obtained at $E = E^{rev}$ on the different electrode materials in $MgCl_2$ as well as the corresponding exchange current densities, are given in Table 1. The activation energy ($\Delta E^*$), determined from the temperature dependency of the cathodic rate constant, is also included in the table.

The cathodic charge transfer coeffcient ($\alpha_c$) was calculated from the potential dependency of the rate constant, and was in general found to be small (0.1–0.3). This is believed to be connected to the phase formation on the electrode. Electrode reactions of metallic electrodes have been reported to have low $\alpha_c$ values (about 0.1 [16]), which is probably due to an unsymmetrical reaction [17]. This unsymmetry is caused by the greater work required to transfer the reactant (Mg(II)-species) than to the transfer of the product (Mg) to the reaction site. The latter is by nature located at the electrode surface and is readily oxidised. In some measurements using the iron electrode the charge transfer coefficient was found to have high values, which means values around 0.5. This was observed in the beginning of the experiment, and at the lower temperatures as the experimental temperature was stepwise increased during the measurements. The recorded change in value of $\alpha_c$ going from 725°C to 780°C is therefore believed to be due to changes of the interfacial properties of the electrode with time rather than a temperature effect. The low values for $\alpha_c$ at 725°C and 735°C can accordingly be understood as if a pure charge transfer (Mg(II)→Mg(0)) took place without any Mg phase present at the electrode.

Also in the chloride–fluoride mixture the charge transfer coefficient was found to be small (about 0.1).

In order to calculate the rate constant from the potential dependent $i_{t=0}$ data the concentration of the electroactive species must be known. For single molten salts (*eg* $MgCl_2$) the concentration of Mg(II) species was calculated by dividing the density of the salt with the molar mass. No literature data of the density of the $MgCl_2$–$MgF_2$ mixture is available. If one assumes that no complex formation takes place, the density can be estimated from the density of the pure components to be about 1.9 g/cm$^3$. The concentration of Mg(II)-species was then calculated to be about $2.2 \times 10^{-2}$ mol/cm$^3$. The obtained values of $i_{t=0}$ ($i_0$) for the $MgCl_2$–$MgF_2$ system, as well as the calculated cathodic rate constants using the calculated values for $c_{Mg(II)}$, are given in Table 2 together with the activation energies from the temperature variation.

The values for the exchange current densities and the rate constant of the charge transfer reaction obtained in this work are smaller than values obtained in more dilute solutions [1], where the cathodic rate constant of the discharge process on a Mg–Ni electrode was calculated to be about

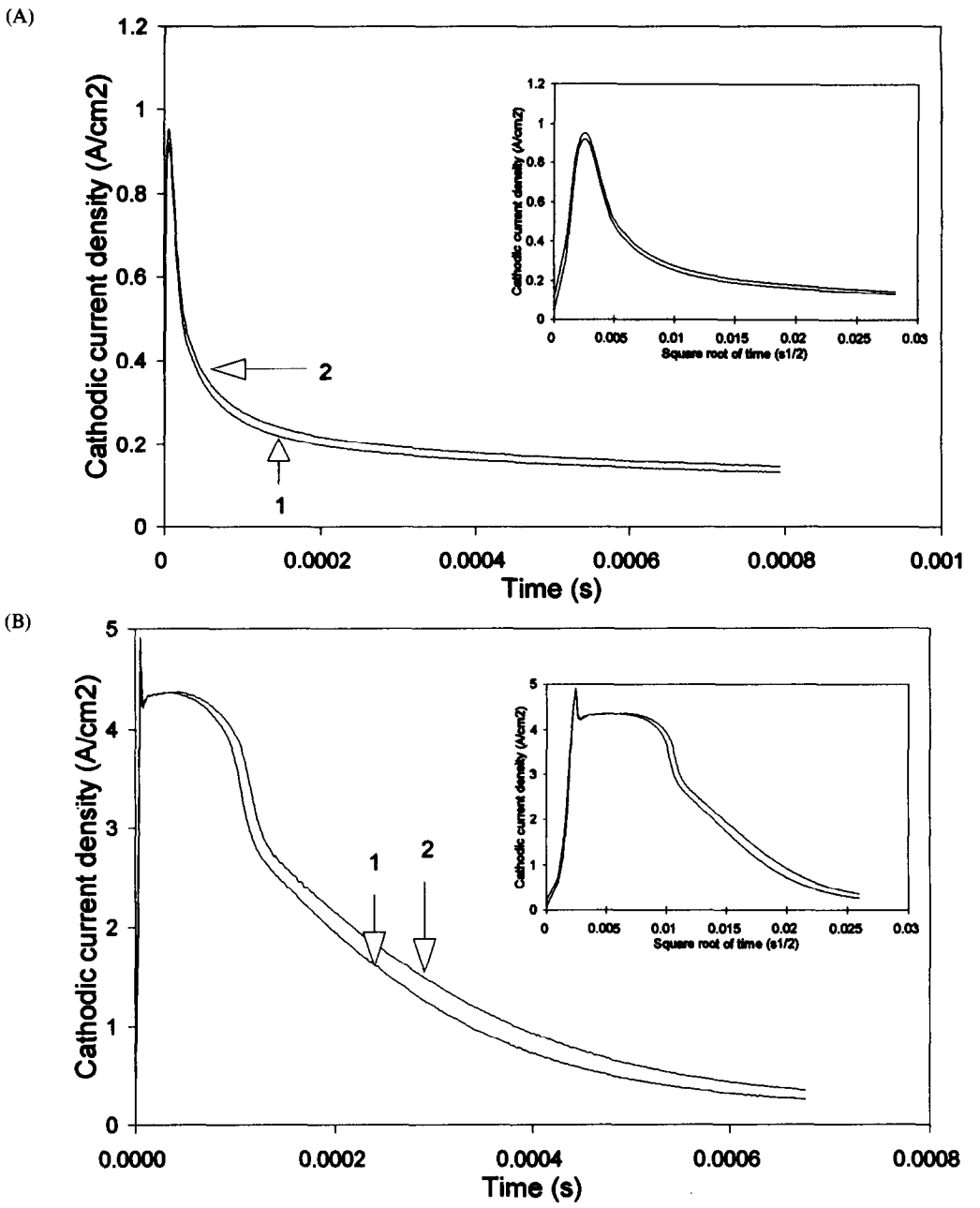



Fig. 2. Cathodic current transients obtained on glassy carbon in (A) $MgCl_2$ at 725°C and (B) $MgCl_2$–$MgF_2$ melts at 656°C. The applied potential in both figures was (1) +20 mV and (2) +10 mV vs magnesium. The transients plotted vs $t^{1/2}$ are given in the inserted figures.

0.01 cm/s. The reason for this discrepancy is not clear, but could be connected to the complexity of the pure $MgCl_2$, implying that the real concentration of the electroactive species is considerably smaller than has been assumed in the calculations. In that case the values both of the exchange cd and the rate constant would accordingly be higher. This has also been shown in recent measurements with *ac*-impedance spectroscopy [18]. Comparing the values reported here with data obtained by impedance measurements on a magnesium electrode [6] shows that the values obtained on the molybdenum electrode are in agreement with the values obtained on the magnesium electrode. This may be understood as if the molybdenum electrode is covered by magnesium, and hence acts as an Mg electrode.

By applying a cathodic potential pulse a hump due to phase formation was observed in Fig. 2(B). The charge associated with this hump was calculated by numerical integration of the current with time. The charge density was found to be 0.5–0.6 mC/cm$^2$ at temperatures above the melting point of magnesium and 0.1–0.3 mC/cm$^2$ at 647°C, which is below the melting point of Mg (650°C). The theoretical charge

Table 1.
Cathodic rate constants ($k_c$) at $E = E^{rev}$, exchange current densities,* charge transfer coefficients, and activation energies obtained on glassy carbon (GC), molybdenum (Mo) and iron (Fe) electrodes in molten $MgCl_2$ at the reversible $Mg^{2+}/Mg$ potential obtained from potentiostatic transients

| Substrate | Parameter | $T = 725°C$ | $T = 735°C$ | $T = 750°C$ | $T = 780°C$ | $\Delta E^a$ (kJ mol) |
|---|---|---|---|---|---|---|
| GC | $k_c \cdot 10^4$ (cm/sec) | 4 | — | 5 | 7 | 85 ± 5 |
| | $i_0$ (A/cm$^2$) | 1.4 | — | 1.7 | 2.4 | |
| | $\alpha_c$ | 0.2 | — | 0.3 | 0.2 | |
| Mo | $k_c \cdot 10^4$ (cm/sec) | 7 | 8 | 9 | 12 | 93 ± 3 |
| | $i_0$ (A/cm$^2$) | 2.3 | 2.6 | 3.0 | 4.2 | |
| | $\alpha_c$ | 0.1 | 0.1 | 0.1 | 0.1 | |
| Fe | $k_c \cdot 10^4$ (cm/sec) | 13 | 13 | 18 | 21 | 83 ± 6 |
| | $i_0$ (A/cm$^2$) | 4.3 | 4.6 | 6.4 | 7.1 | |
| | $\alpha_c$ | 0.6 | 0.5 | 0.2 | 0.2 | |

*The exchange cd is $i_{t=0}$ extrapolated to $E = E^{rev}$ from equation (1).

required to form a dense monolayer of magnesium atoms is about 0.8 mC/cm$^2$. Consequently, the achieved data indicate that the electrode becomes only partly covered by this layer during the initial period (~0.15 ms). By comparing this information with the estimated charge from corresponding peaks in the voltammogram, it seems that the layer formation takes place on a more extended time scale than indicated by the hump in the current transients. Furthermore, the charge density was calculated by integration of the total current, without correcting for any other parallel processes, yielding a too high charge. The charge of this layer was not found to depend significantly on the applied potential in the region studied (+50 mV to −30 mV vs Mg). Similar values have been obtained on iron electrodes in the pure $MgCl_2$ system. To study this phase formation further current transients were plotted as log $i$ vs log $t$. The slopes of these plots were expected to be close to one which would indicate the nucleation of a two dimensional layer [14]. The calculated slopes however were much lower (0.01–0.05) for both the iron electrode in $MgCl_2$ as well as the glassy carbon electrode in the $MgCl_2$–$MgF_2$ mixture. This could be explained by several processes being significant in this time regime. These processes are charge transfer followed by nucleation at the surface or diffusion of dissolved magnesium away from the electrode surface.

Table 2.
Kinetic data obtained on GC in molten $MgCl_2$–$MgF_2$ (78–22 mol%) mixture at the reversible potential

| Temperature (°C) | $i_0$ (A/cm$^2$) | $k_c \times 10^4$ (cm/sec) | $\Delta E^a$ (kJ/mol) |
|---|---|---|---|
| 647 | 4.8 | 11 | 73 ± 4 |
| 656 | 5.5 | 13 | |
| 682 | 6.7 | 16 | |
| 721 | 10.0 | 24 | |

*As in Table 1.

The formation of an adsorbed layer is also most likely the cause of the recorded increase of the apparent electrode capacitance with increase of the imposed current density (cd) during galvanostatic step measurements. The electrode capacitance was determined by galvanostatic potential transients from the potential/time gradient recorded with 1 μs resolution. The capacitance at low cds was found to be about 10 μF cm$^2$ as shown in Fig. 3. This is a value comparable with what is found in other systems [19]. An interesting feature observed is that the dependency of the cd was dramatically reduced by addition of minor amounts of NaCl as shown in Fig. 3. The effect of the applied current density on the measured electrode capacitance was assigned to the formation of an adsorbed layer on the electrode. The influence of addition of sodium chloride would then expectedly affect this layer formation. It was also observed that the voltammetric prepeak became less pronounced when NaCl was added. At 12 wt% NaCl the cathodic prepeak could not be observed at all, indicating that the underpotential deposition was considerably reduced by the addition. This could be due to a small sodium deposition and intercalation, affecting the interfacial properties.

*Magnesium electrodes.* Cathodic current transients obtained by potential steps from the zero current potential on magnesium electrodes exhibited a feature different from the transients discussed so far, as can be seen in Fig. 4(A). The slow initial rise of the current, during approximately 100 μs after applying the cathodic potential, indicates that phase formation takes place also here. The time dependency of this current increase was calculated from a logarithmic plot as d log $i$/d log $t$, and found to be close to 1 (varying from 0.85 to 0.94), indicating nucleation of a two-dimensional layer. It is surprising that such a process takes place even at the magnesium electrode. This might be explained by a very thin oxide layer present at the magnesium surface. The new magnesium layer is hence formed on top of this oxide layer, which probably was formed by

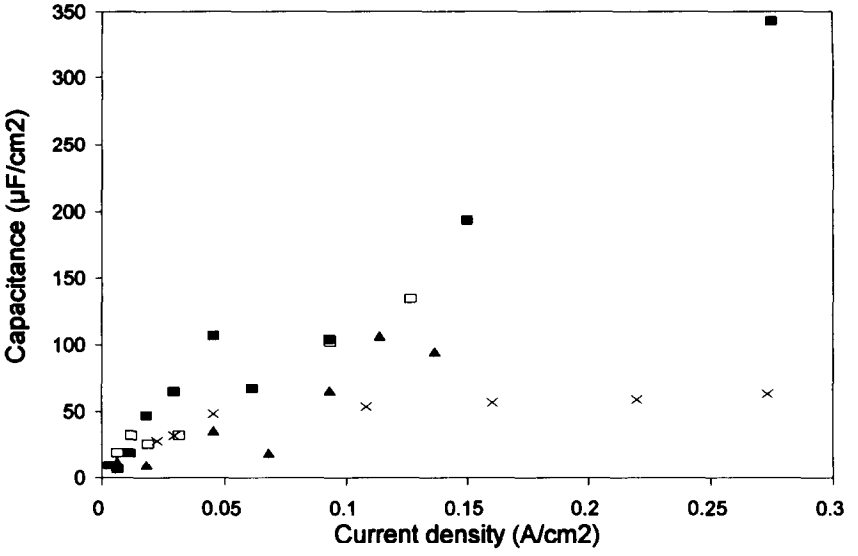



Fig. 3. Electrode capacitance vs applied current density obtained on glassy carbon (■), molybdenum (□) and iron (▲) in molten $MgCl_2$ and on glassy carbon (×) in $MgCl_2$ with 12 wt% NaCl.

reduction of magnesium hydroxychloride and a subsequent precipitation of oxide on the electrode according to equation (2).

$$MgOH^+ + e^- = MgO(s) + 1/2H_2(g). \quad (2)$$

Such a passivation process has previously been observed in our laboratory [20]. This oxide layer is stable in a melt saturated with oxide but will dissolve in a subsaturated melt. The charge associated with this nucleation process was calculated and found to be only a fraction of the charge density calculated on other electrodes, and increasing with the applied potential from about 3 to about 50% of a dense layer. This probably means that nucleation takes place on only a minor part of the surface where the oxide film is thin. The transients obtained on the magnesium electrode were apparently not suited for determination of the rate constant by the extrapolation method used for the other electrodes. The reason for this is that the phase formation obscure the very initial part of the transients where information on the charge transfer kinetics can be obtained directly. For that reason a chronocoulometric method was applied. The cathodic current transients were numerically integrated with respect to time and the corresponding charge was plotted as a function of $t^{1/2}$. Plots of the charge vs $t^{1/2}$ obtained from the transients given in Fig. 4(A) are given in Fig. 4(B). At long times the calculated charge could apparently be described by the following equation [14]:

$$q = 4\,n\,F\,k_c c_b t_L^{1/2}(t^{1/2} - t_L^{1/2})/\pi, \quad (3)$$

where $t_L^{1/2}$ is the intercept on the time-axis in a $q$ vs $t^{1/2}$ plot. The potential dependent rate constant ($k_c$) can hence, according to equation (3), be obtained by calculating the slope of the line and the value of the intercept at zero charge. The calculated rate constants, $k_c$, obtained at $E = E^{rev}$, the corresponding values of $i_0$ and of the activation energy, are given in Table 3. The apparent exchange current density was found to be 4.3 A/cm$^2$ at 720°C in $MgCl_2$–$MgF_2$. Kisza *et al.* [6] calculated a value of 2.15 A/cm$^2$ for the exchange cd for the slowest one-electron step in the process at 720°C in $MgCl_2$. This is exactly the same value if one keeps in mind that the total process is a two-electron transfer reaction.

The activation energy for this electrode is about half the value compared with the other electrodes. Also the rate constant is surprisingly low compared with the values for the glassy carbon electrode. The reason for the low value of the cathodic rate constant on the magnesium electrode is not fully understood, but could be related to a small active electrode area. A possible explanation is that the electrode area for the magnesium electrode was assumed to be the inner cross-section of the ceramic tube containing the magnesium phase. If the active electrode area was smaller, the real rate constant would consequently be higher. Estimation of the limits of the electrode area indicates, however, that the uncertainty in area determination cannot explain the low value. A more reasonable explanation is that the electrode was slightly passivated by oxide. It has been reported above that a nucleation process, most likely on a thin oxide layer, has been observed on magnesium. For that reason it is believed that this oxide layer, to a certain degree, passivated the electrode and led to a reduced rate of the charge transfer. The charge transfer coefficient for the Mg electrode was found to be about 0.6 at all temperatures. This was rather surprising since, according to the previous discussion, one would expect a low value for $\alpha_c$ on a metallic Mg electrode. However, if the electrode was covered by an oxide layer, this probably inhibited the oxidation

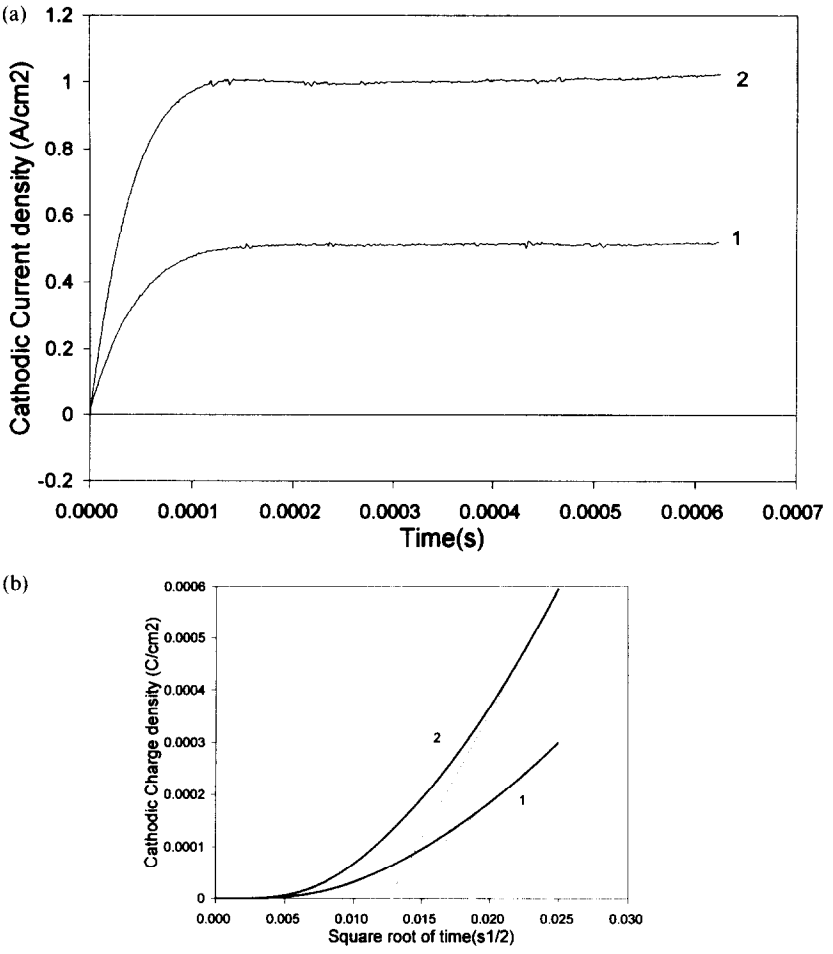



Fig. 4. (a) Current transients obtained on an $Mg_{(l)}$ electrode in $MgCl_2$–$MgF_2$ mixture at 699°C. The applied potential was (1) −10 mV and (2) −20 mV vs $Mg^{2+}$/Mg. (b) Corresponding charge density ($q$) plotted vs $t^{1/2}$.

of the Mg phase and the substrate acted as an inert ("heterogeneous") electrode.

### Diffusion

At potentials positive to the critical value for nucleation a diffusion-controlled process has been observed. Due to the fact that these transients were obtained in systems where Mg(II)-species are the only cations the mass transfer-controlled process is most likely diffusion of the product (dissolved metal) from the electrode interface to the electrolyte. A pure diffusion-controlled process should obey a linear $i$ *vs* $t^{-1/2}$ relationship (Cottrell equation). As observed in Fig. 5 a linear relationship is observed in an intermediate period (0.2–1 sec). At longer times the points deviate from the linear behaviour due to underpotential two dimensional layer formation (UPD) and/or convection in the melt. From the slope of the lines in the linear region, in plots similar to those shown in Fig. 5, the interfacial concentration of the diffusing species can be calculated if the diffusion coefficient is known. Van Norman and Egan [21] have reported a value of the diffusion coefficient for dissolved Mg equal to $5.3 \times 10^{-6}$ cm$^2$/sec.

Table 3.
Kinetic data obtained on an Mg electrode in molten $MgCl_2$–$MgF_2$. The symbols have the same meaning as in Table 1

| Temperature (°C) | $k_c c \times 10^5$ (mol sec$^{-1}$ cm$^{-2}$) | $k_c \times 10^4$ (cm/s) | $i_o$ (A/cm$^2$) | $\Delta E^a$ (kJ/mol) |
|---|---|---|---|---|
| 647 | 1.47 | 6.6 | 2.8 | This work: 44 ± 4 |
| 660 | 1.62 | 7.4 | 3.1 | $MgCl_2$: 60.6 [6] |
| 699 | 2.09 | 9.5 | 4.0 | |
| 720 | 2.22 | 10.0 | 4.3 | |

[a]As in Table 1.

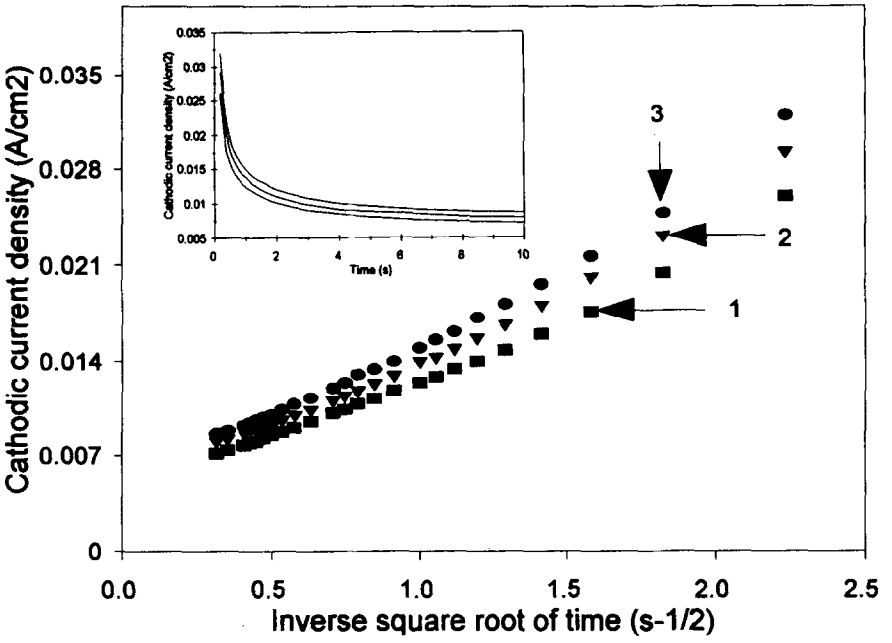



Fig. 5. Cathodic current transients obtained on glassy carbon in $MgCl_2$–$MgF_2$ at 668°C plotted vs $t^{1/2}$. The applied potential was (1) +20 mV, (2) +10 mV and (3) 0 mV vs $Mg^{2+}/Mg$.

However these authors used a value of the equilibrium solubility of magnesium equal to 0.23 mol%, which is probably too high. Substituting the metal solubility reported by Wypartowicz *et al.* [22] (0.14 mol%) to the values reported by Van Norman and Egan yields a diffusion coefficient equal to $1.43 \times 10^{-5}$ cm$^2$/s. By using this value of the diffusion coefficient the concentration of the diffusing species was calculated to be in the range of $10^{-5}$ mol/cm$^3$. Some values achieved on a glassy carbon electrode in molten $MgCl_2$ are collected in Table 4.

Table 4.
Concentration of diffusing species calculated from potentiostatic current transients obtained on glassy carbon in $MgCl_2$ and $MgCl_2$–$MgF_2$. The theoretical values calculated from the Nernst equation based on equilibrium solubilities, are given in parentheses

| Temperature (°C) | Potential (mV) | Concentration × $10^5$ (mol/cm$^3$) |
|---|---|---|
| $MgCl_2$ | | |
| 725 | +20 | 4.2 (1.4) |
| | +10 | 4.3 (1.8) |
| | 0 | 4.4 (2.3) |
| 750 | +20 | 4.7 (1.5) |
| | +10 | 5.2 (1.9) |
| | 0 | 6.0 (2.5) |
| 780 | +20 | 7.6 (2.3) |
| | +10 | 6.6 (2.9) |
| | 0 | 8.7 (3.6) |
| $MgCl_2$–$MgF_2$ | | |
| 647 (solid) | +20 | 1.3 |
| | +10 | 1.6 |
| | 0 | 1.9 |
| 668 (molten) | +20 | 2.1 |
| | +10 | 2.4 |
| | 0 | 2.6 |
| 685 (molten) | +20 | 3.7 |
| | +10 | 4.0 |
| | 0 | 5.0 |

The values obtained at $E = E^{rev}$ ($\eta = 0$ mV) should be equal to the equilibrium concentration of magnesium dissolved in the melt. As can be observed in Table 4 the calculated values are about twice the values given by Wypartowicz *et al.* [22]. Similar results have also been obtained on other electrode materials, indicating that the electrode process at these potentials is not very dependent on the substrate. From the table one may notice that the determined values of the concentration generally increase with an increase of the cathodic potential and the temperature (except for +10 mV at 780°C), a behaviour which agrees with the perception that the cathodic process is the formation of dissolved metal.

The value of the diffusion coefficient of dissolved metal in the chloride–fluoride mixture has not been reported, to the knowledge of the authors. If one assumes that the value of the diffusion coefficient in the two systems is the same, the metal solubility is apparently slightly higher in the latter system, as observed by comparing the values in $MgCl_2$–$MgF_2$ at 685°C and in $MgCl_2$ at 725°C in Table 4. However, the difference in the values obtained in the pure $MgCl_2$ compared to the chloride–fluoride mixture, could also be due to the different levels of underpotential nucleation (UPD) as shown in Fig. 2(A) and 2(B).

Another way to study the behaviour of the product is to oxidise it. This was carried out in the following (less accurate) way; the electrode was first polarised

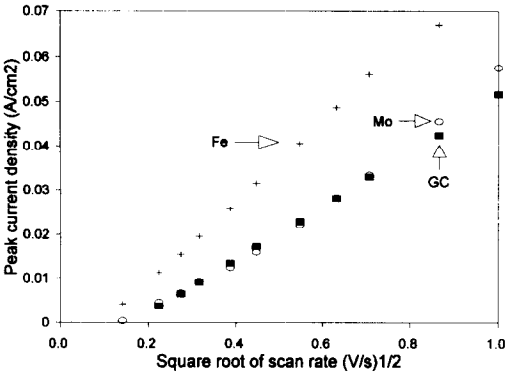



Fig. 6. Peak current density vs the square root of sweep rate ($v^{1/2}$) obtained on glassy carbon, iron and molybdenum electrodes in molten $MgCl_2$ at 725°C. The applied potential was +10 mV vs $Mg^{2+}/Mg$.

cathodically to the zero current potential, but positive to nucleation, at constant potential for a certain time (20 sec). The purpose of this polarisation period was to fix the concentration of the electrode products at the electrode interface. The potential was then changed anodically and linearly with time at varying sweep rates, resulting in an anodic peak current due to oxidation of the electrode products. The peak cd vs $v^{1/2}$ obtained on different electrode materials in $MgCl_2$ is given in Fig. 6. From the slope of these curves the concentration of the species being oxidised has been calculated according to equation (4) [14].

$$i_p = 0.4463\,((nF)^{3/2}/(RT)^{1/2})\,c^{\infty}D^{1/2}v^{1/2}. \qquad (4)$$

In accordance with the data obtained by potential step measurements the concentration increases with a decrease in the applied potential and was found to be about $5 \times 10^{-5}$ mol/cm$^3$ at 725°C when the polarisation potential was zero mV. In general the values obtained from potential step current transients, using the Cottrell equation, and from oxidation of dissolved metal, were comparable and somewhat high compared to the literature value, which is about $2 \times 10^{-5}$ mol/cm$^3$. Considering the voltammetric oxidation of the *in situ* formed Mg dissolved in the melt, one has assumed that the transient diffusion layer created by the anodic sweep is much thinner than the steady state diffusion layer formed by the primary cathodic polarisation. Thus the apparent "bulk" concentration of dissolved metal is rather constant during the sweep and close to the concentration at the electrode interface during the polarisation period.

The concentration of dissolved species at the interface is determined by the applied potential which is more positive than, or equal to, the equilibrium potential for an Mg phase. This concentration cannot then be higher than the concentration of dissolved Mg in equilibrium with such an Mg phase. From this reasoning the approximate use of equation (4) cannot explain the observed deviation, since we calculate a higher concentration than the reported equilibrium solubility. The use of unsheathed electrodes will involve an uncertainty in the determination of the electrode area and hence the current density. Estimations indicate that the uncertainty related to the determination of the electrode area is about 10%, which cannot explain the deviation in the concentration of dissolved Mg. Another possible explanation is the reduction of impurities dissolved in the melt. The most likely impurity in this respect is magnesium hydroxychloride (MgOHCl). The melt was analysed and the hydroxychloride concentration was estimated to be about 0.02 mol% ($\sim 4 \times 10^{-6}$ mol/cm$^3$). Calculations have shown that this level of hydroxychloride neither can explain the observed deviation. The most reasonable explanation is therefore the oxidation of a mono- or submono layer formed in parallel with the formation of dissolved metal at the electrode. Another possibility of course is that the reported values of the diffusion coefficient and/or the equilibrium solubility are not fully correct.

## CONCLUSIONS

The measurements of the rate of the charge transfer has been found to be influenced by the formation of an adsorbed layer. This underpotential deposition is most likely the formation of a two dimensional layer of magnesium. Such underpotential deposition has been observed on electrodes of glassy carbon, iron, molybdenum and even on magnesium. The observed nucleation process on the magnesium electrode most likely shows that a, probably very thin, oxide layer is formed on magnesium metal at these conditions.

The apparent cathodic rate constant has been found to have a value of about $10^{-3}$ cm/s, ranging from $0.4 \times 10^{-3}$ on GC, $0.7 \times 10^{-3}$ on Mo and $1.3 \times 10^{-3}$ on Fe, at 725°C in $MgCl_2$, to $1.0 \times 10^{-3}$ on molten Mg at 721°C in $MgCl_2$–$MgF_2$. The rather low rate constant on molten Mg is probably due to a partial passivation of the Mg-surface. The activation energy determined on the different foreign electrode materials in $MgCl_2$ was found to vary between 83 and 93 kJ/mol. The corresponding values obtained on GC and Mg in $MgCl_2$–$MgF_2$ were 73 kJ/mol and 44 kJ/mol, respectively.

The diffusion process observed at potentials positive to the nucleation potential has been attributed to the formation of dissolved metal, which diffuses from the electrode interface into the electrolyte. The current associated with the formation of homogeneously dissolved metal will increase with increasing cathodic potential. The formation of dissolved metal will hence be a source of reduced current efficiency in industrial electrolysis cells.

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